Module 2

Lasers & Fiber Optics

Introduction, Interaction of radiation with matter, expression for energy density at thermal equilibrium in terms of Einstein's coefficient, conditions and requisites of Laser, characteristics of laser, Quantum Dot Laser and their applications, Problems.

Principle and propagation of light in optical fibers, Numerical aperture and Acceptance Angle, Types of Optical fibers, Attenuation, application- point to point telecommunication network, Problems.

Extra Learning: Application of Lasers & Optical fiber endoscopy

Introduction:

LASER is an acronym for Light Amplification by Stimulated Emission of Radiation. Laser produces narrow, amplified, coherent beam of light. Production of laser is as a result of interaction of radiation with matter. First laser was built by Theodore H. Maiman in 1960

Differences between ordinary light and Laser beam.

S.No	Property	Ordinary light	Laser beams
1	Angular spread	In ordinary light the angular spread is more	In laser beam the angular spread is less.
2	Directionality	They are not directional	They are highly directional
3	Intensity	It is less intense	It is highly intense
4	Coherence	It is not a coherent beam and is not in phase	It is a coherent beam and is in phase
5	wavelength	The radiation are polychromatic	The radiations are monochromatic
6	Example	Sun light, Mercury vapor lamp	He- Ne Laser, Co2 laser

Interaction of radiation with matter:

The absorption and excitation of atoms to higher energy states, emission of energy by matter and the resulting de excitation of atoms to the lower energy states are as a result of the interaction between matter and radiation which forms the basis of emission of Laser. It is seen that, the Interaction of matter and radiation can happen in three ways, which are as listed.

i) Induced Absorption
ii) Spontaneous emission &
iii) Stimulated Emission.

Let us discuss them in detail. Consider a system with two energy states E_1 (lower state) and E_2 (higher state)

Induced Absorption: A system in a lower state E1 absorbs radiation and going to a higher state E_2 is called induced absorption. This can happen if and only if the energy of the radiation is at least equal to the difference between the energy levels. This can be represented by the equation,

$$h\nu = E_2 - E_1.$$

This interaction can be represented as, Atom + Photon → atom*

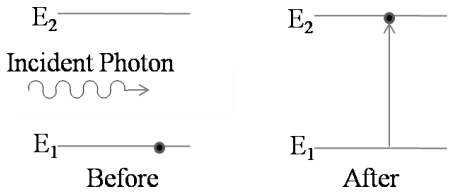


Figure 2.1 Induced Absorption

The process of induced absorption is represented diagrammatically in figure 2.1

Spontaneous Emission: A system at a higher energy level E_2 comes to the lower energy level E_1 by emitting radiation of energy,

$$h\nu = E_2 - E_1,$$

by itself without the aid of any external agency is called spontaneous emission. The process of spontaneous emission is represented in figure 2.2

Atom* → Atom + Photon

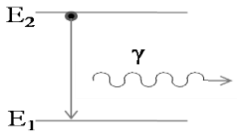


Figure 2.2 Spontaneous Emission

Stimulated Emission: A system at a higher energy level E_2 comes to the lower energy level E_1, by the emission of radiation of energy

$$E = h\nu = E_2 - E_1,$$

if and only if another photon of same energy is passed by. There are 2 emitted photons, both (in same phase, λ, amplitude) coherent. One is called stimulating Photon & other stimulated photon.

Atom* + Photon → Atom + 2 Photons

Figure 2.3 represents the process of stimulated emission.

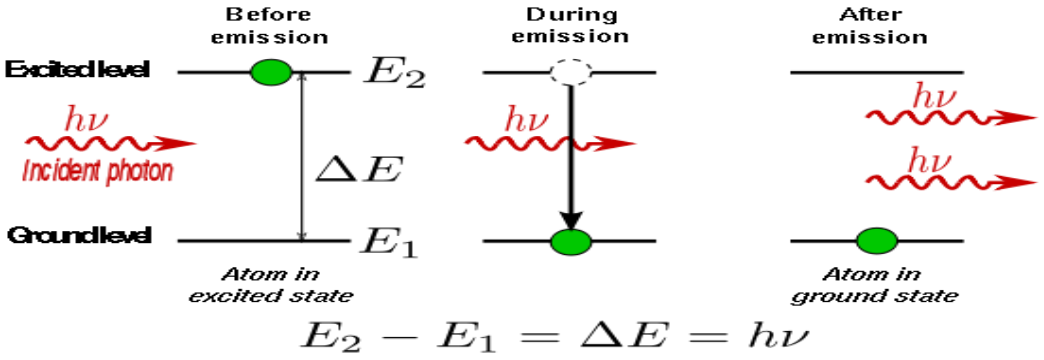


Figure 2.3 Stimulated Emission

Expression for Energy Density at thermal equilibrium in terms of Einstein's coefficients

Einstein's coefficients:

Einstein obtained an expression for energy density at thermal equilibrium of a system, in terms of important parameters known as Einstein's coefficients.

Consider two energy states E_1 and E_2 of a system. Let, $E_2 > E_1$
N_1 → no. of atoms / unit volume in state E_1
N_2 → no. of atoms / unit volume in state E_2
U_ν → Energy density of the radiation of frequency ν
The Einstein's coefficient for the three processes are obtained as follows.

Induced Absorption: Here the atoms at level E_1 goes to E_2 by absorbing radiation of energy

$$h\nu = E_2 - E_1 \; : \; \nu = \frac{E_2 - E_1}{h}$$

The rate of induced absorption is proportional to,

i) No. of atoms/unit volume in state E_1 &
ii) Energy density of incident radiation U_ν

$$\therefore \text{Rate of induced absorption } \alpha \; N_1 U_\nu$$

$$Rate\ of\ induced\ absorption = B_{12} N_1 U_\nu \ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Where,
B_{12} → Einstein's coefficient of induced Absorption or probability of induced absorption.

Spontaneous emission: Here the atoms at level E_2, come to E_1, by itself, by emitting radiation of energy

$$E = E_2 - E_1 \quad \text{or } \nu = \frac{E_2 - E_1}{h}$$

The rate of spontaneous emission is proportional to the number of atoms/unit volume in level E_2.
Rate of spontaneous emission α N_2

Rate of spontaneous emission = $A_{21}N_2$ ……..……(2)

A_{21}→Einstein's coefficient of spontaneous emission or probability of spontaneous emission.

Stimulated Emission: Here, the atoms at level E_2 comes to state E_1, by emitting photon of energy $E = E_2-E_1$, only if it is aided by another photon of same energy. The rate of stimulated emission is proportional to

i) No. of atoms/unit volume in state E_1,

ii) Energy density of the radiation U_ν

Rate of stimulated Emission α N_2U_ν

Rate of stimulated Emission = $B_{21}N_2U\nu$ ……(3)

B_{21} → Einstein's coefficient of stimulated Emission

At thermal Equilibrium,

Rate of absorption = Rate of emission

∴ Rate of induced absorption = Rate of spontaneous emission + Rate of stimulated emission

……...(4)

Substituting (1), (2) & (3) in (4)

$B_{12}N_1U_\nu = A_{21}N_2 + B_{21}N_2U_\nu$

$B_{12}N_1U_\nu - B_{21}N_2U_\nu = A_{21}N_2$

$(B_{12}N_1 - B_{21}N_2)U_\nu = A_{21}N_2$

$U_\nu = \frac{A_{21}N_2}{(B_{12}N_1-B_{21}N_2)} = \frac{A_{21}N_2}{B_{21}N_2\left(\frac{B_{12}}{B_{21}}\frac{N_1}{N_2}-1\right)}$ ---------- (5)

From Boltzmann's theorem

$N_1 / N_2 = e^{h\nu/kT}$----------(6)

Substituting (6) in (5)

$U_\nu = \frac{A_{21}}{B_{21}\left(\frac{B_{12}}{B_{21}}e^{h\nu/KT}-1\right)}$

$$U_\upsilon = \frac{A_{21}}{B_{21}}\left[\frac{1}{\left(\frac{B_{12}}{B_{21}}e^{\left(\frac{h\upsilon}{kT}\right)}-1\right)}\right] \quad \text{…….……(7)}$$

According to Planck's law, $U_\nu = 8\pi h\frac{\nu^3}{c^3}\{\frac{1}{\left(e^{\frac{h\nu}{KT}}-1\right)}\}$ ……..…… (8)

Comparing (7) & (8) , $\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{C^3}$

$$\frac{B_{12}}{B_{21}} = 1 \qquad \therefore \; B_{12} = B_{21}$$

Therefore,

Einstein's coefficient for induced absorption = Einstein's coefficient for stimulated emission.

In general, $U_\nu = \frac{A}{B(e^{h\nu/KT}-1)}$ ---------- (9)

i) $$\frac{stimulated\ emission}{Absorption} = \frac{B_{21}N_2U_\nu}{B_{21}N_1U_\nu} = \left(\frac{B_{21}}{B_{12}}\right)\frac{N_2}{N_1}$$

$$= \frac{N_2}{N_1} \qquad (B_{21} = B_{12})$$

Amplification is high only when $N_2 > N_1$

Therefore, population inversion is must for laser action.

Condition and requisites for Laser action: - The process of stimulated emission dominates only when the population of the higher energy state (E_2) is higher than the population of the lower energy (E_1). This condition is called population inversion. The stimulated emission results in the emission of stimulated photons. This is called laser emission.

For laser action to take place the requisites are,

i) Optical resonant cavity or Laser cavity
ii) Active medium
iii) Pumping

The conditions for Lasing action to take place are,

i) Population inversion
ii) Stimulated Emission
iii) Optical feedback
iv) Metastable state

Laser cavity: It converts the active medium into light generator. It consists of the active medium bound between two mirrors. (The active medium together with the two mirrors form optical cavity). One mirror is 100% reflecting and the other is 90% reflecting. The two mirrors along with the active medium form the cavity, in which two types of waves, one moving towards right and the other moving towards left exist. The length of the optical cavity should be designed in such a way that, path length travelled by the wave 2L (L→ length of optical cavity) should support constructive interference.

i.e. $2L = m\lambda$ or $L = \frac{m\lambda}{2}$

Active medium: A medium in which light gets amplified. The medium can be solid, liquid or gas. Only a small fraction of the medium is responsible for stimulated emission and amplification and they are called active centres. The remaining part of the medium supports the active centres.

Pumping: The process of exciting the atoms to the required higher energy level is called pumping. If pumping is done by optical energy, it is called optical pumping. [eg.: Ruby laser]. The pumping done by electrical energy is called electrical pumping. [eg.: He-Ne laser]. If pumping is done by chemical reactions, it is called chemical pumping.

Metastable state: Any ordinary excited state cannot be used to achieve the condition for population inversion. Metastable state is a special excited state, where an atom can stay for more than a milli second (10^{-3} sec). But in an ordinary excited state, an atom can stay for less than 10^{-9} sec (nano sec). The metastable state helps to attain the population inversion.

Population inversion: Normally, the population of atoms in lower energy state is always greater than the population of atoms in higher energy state. Consider three energy state E_1, E_2 & E_3, where $E_1 < E_2 < E_3$. Let E_2 be the metastable state. By supplying suitable energy let the system goes from E_1 to E_3. E_3 being an ordinary excited state, atoms get de-excited from E_3 toE_2 or E_1 as represented in figure 3.4

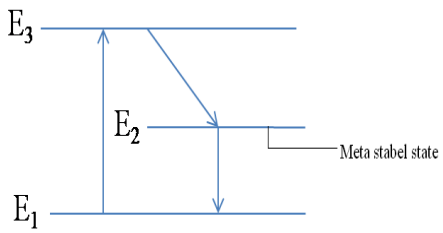


Figure 3.4 Population Inversion

At E_2, atoms can stay for more than millisecond. So the population of atoms at E_2 increases rapidly and becomes greater than the population of E_1. This condition, where the population of higher energy state is greater than the lower energy state is called "Population inversion".

Stimulated emission: The process in which an external photon is used to make a system come from its excited state to lower state, thereby emitting two coherent photons is called stimulated emission. Once population inversion is attained stimulated emission can be made to happen by passing a photon of right energy. All the atoms in the metastable state can be brought to ground state, by emitting photons of same energy and phase.

Characteristics of LASER beam:

The main characteristics of Laser beam are,

1. Highly Directional
2. High Intensity
3. Monochromatic
4. Coherent

QUANTUM DOT LASER AND THEIR APPLICATIONS

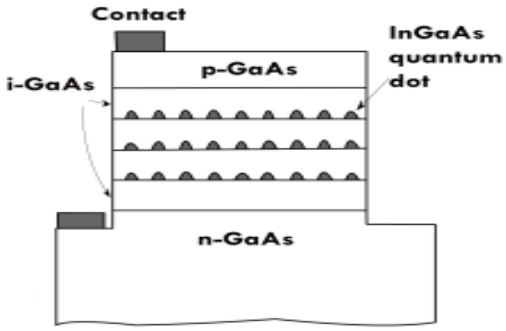
Introduction:

A quantum dot laser is a **semiconductor laser** that uses **quantum dots as the active medium** in its light emitting region. It tightly confines the electrons and holes to produce steady output, regardless of external temperature.

A quantum dot is a non-traditional semiconductor nanoparticle which has 0 degrees of freedom (charges confined in all directions) and behaves as a whole unit in terms of its transport and properties. Their diameter ranges from 2-10 nanometres (consists of 10-50 atom cluster)

Construction:

There are a number of direct band gap semiconductors that can be used for laser devices. Like GaAs based lasers are used in the near IR region, while in the visible region, CdX (X is a chalcogenide) based lasers are used. Laser devices are usually single or double hetero structures. schematic view of the band structure of a typical quantum dot laser is given Fig 1.

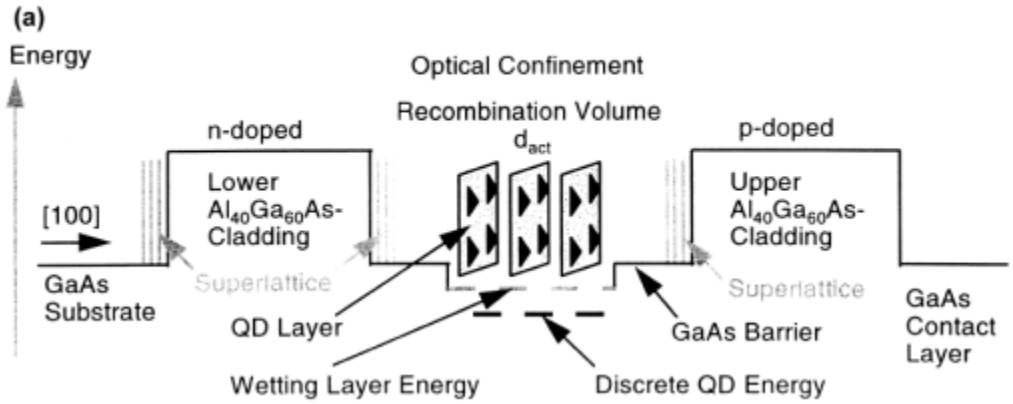


An ideal QD laser consists of a 3D-array of dots with equal size and shape (middle of the figure), surrounded by a higher band-gap material which confines the injected carriers. The thickness of active layer (in all the three directions) in a hetero junction laser can be made very thin comparable to the de Broglie wavelength of the electrons (typically around the nm range) to achieve quantum confinement. The energy levels are no longer continuous but are quantized. The spacing between

the quantized energy levels depend on the width of the active layer. This also controls the wavelength of the laser radiation.

Also the efficiency of a quantum dot laser is greater than that of a bulk laser due to a tailoring of the distribution of electrons and holes that are involved in the stimulated emission (light producing) process.

The whole structure is embedded in an optical waveguide consisting of lower and upper cladding layers (n-doped and p doped shields). Here p-GaAs and n-GaAs is usually AlGaAs layer. The beam is confined to the active GaAs region by total internal reflection at the GaAs-AlGaAs layer. This is because the refractive index of the GaAs is higher than AlGaAs.



Working Principle:

The laser structure consists of three layers of InGaAs quantum dots each of which is grown in a matrix of GaAs (10 nm thick). These are themselves grown in $Al_{0.3}Ga_{0.7}As$, and together comprise the waveguide core of the device. Atomic force microscopy (AFM) studies indicate the dots are lens like in shape, are 2.2nm high and 36 nm in diameter with a dot density of $4.5x10^{10}$ cm $^{-2}$. When a forward bias is applied, carriers are injected into the active region. The difference between quantum dot energy levels can be used for the laser action instead of the band gap. This is very useful since the wavelength of light emitted can be tuned simply by altering the thickness of the layer. These active region carriers which are injected achieves population inversion, and the recombination leads to emission. The emitted light is reflected by the mirrors at the end of the device and the inactive regions, with their lower refractive index, confine the light within the active region. The light intensity is proportional to the forward bias current. There is a minimum forward bias current, called threshold current, I_{th}, above which stimulated emission takes place and here the intensity is directly proportional to the current. At low current, spontaneous emission dominates, the threshold current depends on the laser structure, the optical losses due to absorption, the refractive index of the various layers and the device dimensions. Using different Q-dot structures scientists have achieved emission of light in the ranges of:

– 0.9-1.1 μm
– 1.27-1.3 μm
– 1.9 μm

Advantages of Q-dot lasers:

- Broad range of light emission by changing dot size
- Small active volume means smaller population inversion necessary for lasing.
- Less temperature dependence of threshold current density
- Increase in gain (2-3 times larger than Q-well)
- High frequency operation
- More efficient laser

Challenges

• Advantages depend on dot characteristics
• Challenge is in forming high quality dots 1. Uniform size, 2. Higher density

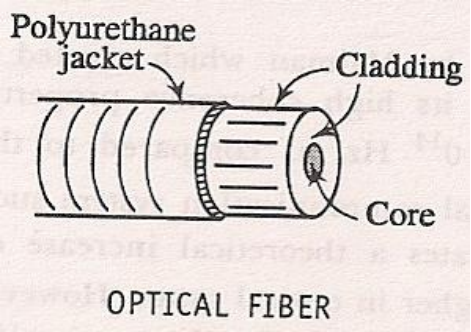
Application of Quantum dot Laser:

Quantum dot lasers find commercial application in medicine (laser scalpel, optical coherence tomography), display technologies (projection, laser TV), spectroscopy and telecommunications. In comparison to the performance of conventional strained quantum-well lasers of the past, the new quantum dot laser achieves significantly higher stability of temperature.

FIBER OPTICS

They are light guides used in optical communication as wave guides. They are made of transparent dielectric materials like glass or plastic.

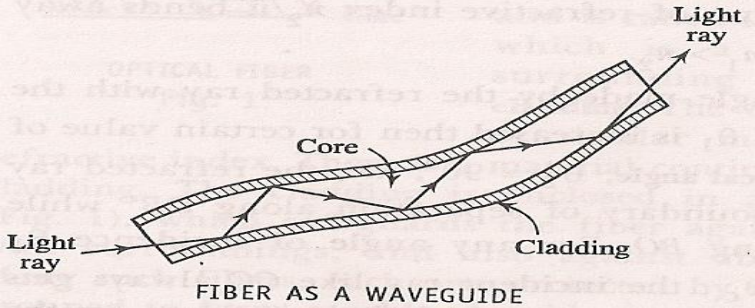
An optical fibre is made of two parts. One is the inner cylindrical material made of glass or plastic called core and the other is called cladding which envelopes the core. Cladding is made of same material as that of core but with lesser refractive index. The cladding envelopes core as concentric cylinder. There is a material continuity between core and cladding. The cladding is enclosed in a polyurethane jacket which safeguards the fibre against any chemical reaction with the surrounding. Many such fibres, each protected by individual jacket grouped to form a cable. A cable may consist of one to several hundred fibres.



OPTICAL FIBER

A waveguide is a tubular structure through which energy can be guided in the form of waves. This is also called light guide (or) fibre wave guide. The guiding mechanism is as follows :

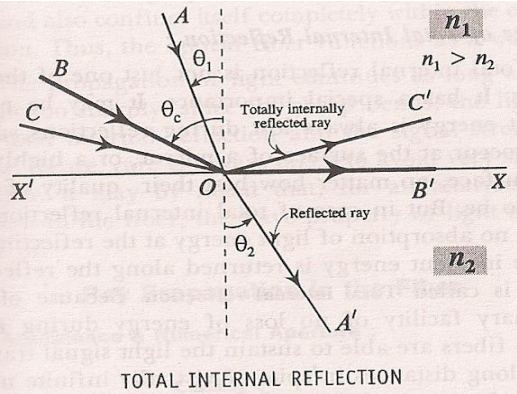
We know that in an optical fibre cladding has a R.I always lesser than that of the core. The light signal which enter into the core can strike the core-cladding interface only at large angles of incidence. The light signal undergoes reflection after reflection with the core. Since each reflection is T.I.R the signal sustains its strength and also present within the core during propagation.



FIBER AS A WAVEGUIDE

The guiding or propagation of light continues as long as the wave guide or fibre is not bent too sharply. This is because for sharp bends the light does not undergo T I R and hence the strength of the signal drops down. Therefore, care should be taken to avoid very sharp bends in the fibre.

The principle mainly involved in optical guide is T I R.



TOTAL INTERNAL REFLECTION

Consider a plane xx' separating two media of R.I n_1 & n_2. Let $n_1 > n_2$. A light wave AO be incident on xx' at O at an angle of incidence θ_1. Since $n_1 > n_2$, the light gets refracted as OA' with θ_2 as the angle of refraction. Also $\theta_2 > \theta_1$. As the angle of incidence θ_1 is increased, angle of refraction θ_2 also increases and for an angle θ_c, the refracted ray just grazes the boundary ,(i.e) $\theta_2 = 90^0$. For any angle of incidence $\theta_1 > \theta_c$, the ray gets reflected back into the medium of R.I n_1. This is called TIR. The angle θ_c, for which $\theta_2 = 90^0$ is called critical angle. According to Snell's law,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

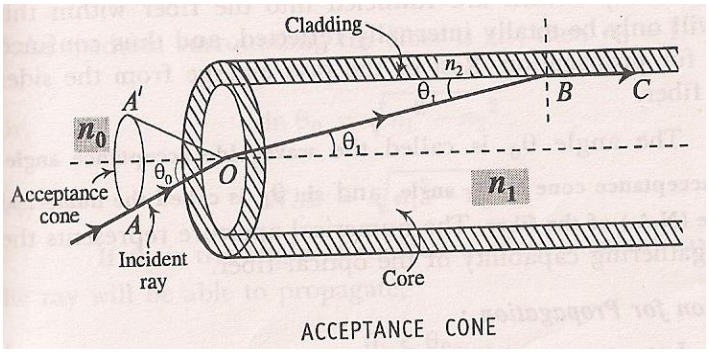
When $\theta_1 = \theta_c$, $\theta_2 = 90^0$,

$n_1 \sin\theta_c = n_2$, $\sin\theta_c = n_2/n_1$

$\theta_c = \sin^{-1}(n_2/n_1)$.

TIR has a special importance. During any other reflection like plane mirror reflection, some energy is lost. But in TIR there is absolutely no loss in energy. Because of this property, optical fibres are able to sustain the light signal transmission over a long distance.

Ray Propagation and Numerical aperture:



Consider the ray AO entering into the core at an angle θ_0 with the axis of the core. Let this gets refracted as OB at an angle θ_1 to the axis. Let the angle of incidence at B be critical angle. Therefore the ray just graces the core – cladding interface as BC. The angle of incidence at B is $(90 - \theta_1)$.

From the fig. it is clear that any ray falling into the core making an angle less than θ_0, will have the angle of refraction less than θ_1. Due to this the angle of incidence at B becomes greater than critical angle and hence the ray undergoes TIR.

On the other hand any ray entering the core with an angle greater than $\theta_0$will be incident at B making an angle less than the critical angle and so gets refracted and lost.

If OA is rotated around the axis of the core by keeping θ_0 same, it is seen that those rays which are funneled into the core within this cone of half angle θ_0 will only be TIR and the rest of the incident light emerge from the sides of the fibre.

Angle θ_0 is called acceptance angle and $\sin\theta_0$ is called numerical aperture. NA represents the light gathering capacity of optical fibre.

Let n_0, n_1, n_2 be the r.I of the surrounding, core and cladding respectively. For refraction of ray A_0

$$n_0 sin\theta_0 = n_1 sin\theta_1 \ \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

At B,

$$n_1 \sin(90 - \theta_1) = \sin 90 * n_2$$

$$n_1 \sin(90 - \theta_1) = n_2$$

$$cos\theta_1 = \frac{n_2}{n_1}$$

Also we have, from eqn.(1)

$$sin\theta_0 = \frac{n_1}{n_0} sin\theta_1$$

$$= \frac{n_1}{n_0}\sqrt{(1 - cos^2\theta_1}$$

$$= \frac{n_1}{n_0}\sqrt{\left(1 - \frac{{n_2}^2}{{n_1}^2}\right)}$$

$$= \frac{n_1}{n_0}\frac{\sqrt{{n_1}^2 - {n_2}^2}}{n_1}$$

$$sin\theta_0 = \frac{\sqrt{{n_1}^2 - {n_2}^2}}{n_0}$$

If the surrounding media in air, $n_0 = 1$

N.A $= \sqrt{{n_1}^2 - {n_2}^2}$

A ray with an angle of incidence θ_1 will be propagated if and only if $\theta_1 < \theta_0$

$$sin\theta_1 < sin\theta_0$$
$$sin\theta_1 < N.A$$

Fractional index change Δ is defined as the ratio between the change in Refractive Index of core and cladding to the refractive index of the core.

$$\Delta = \frac{n_{1-n_2}}{n_1}$$

$$\therefore (n_1 - n_2) = \Delta n_1$$

Also, $N.A = \sqrt{{n_1}^2 - {n_2}^2} = \sqrt{(n_1 + n_2)(n_1 - n_2)}$

N.A = $\sqrt{(n_1 + n_2)\Delta n_1} = \sqrt{2n_1 X \Delta n_1)} = n_1\sqrt{2\Delta}$

N.A = $n_1\sqrt{2\Delta}$

If Δ increases, NA also increases and hence the light gathering capacity of optical fiber increases. But Δ cannot be increased to a very large extent, since it causes signal distortion.

Types of optical fibers

Optical fibers are classified into three major categories

i. The type of material used
ii. The number of modes
iii. The refractive index profile

Based on the type of the material used, they are classified into two types

1. Glass fiber:
   Example: Core: SiO2 Cladding: SiO2
   Core: GeO2- SiO2 Cladding: SiO2
2. Plastic fiber:
   Example: Core: polymethyl methacrylate : Cladding: Co- Polymer
   Core: Polystyrene : Cladding: Methyl methacrylate

Based on the number of modes, they are classified as

1. Single mode fiber
2. Multimode fiber

Based on the refractive index profile, they are classified as

1. Step- index fiber
2. Graded index fiber

Mode : It is defined as the number of rays a fibre can support. If a fibre can support only one ray for propagation, it is called single mode fibre. If a fibre can support more than one mode for propagation it is called Multimode fibre.

In a multimode fibre, the number of modes depend on numerical aperature, diameter of core, wavelength of ray it propagates. The number of modes is given by $N \simeq \frac{V^2}{2}$,

Where V is V-number

V-number = $\frac{\pi d}{\lambda}$ (NA)

V-number = $\frac{\pi d}{\lambda} \frac{\sqrt{{n_1}^2-{n_2}^2}}{n_o}$

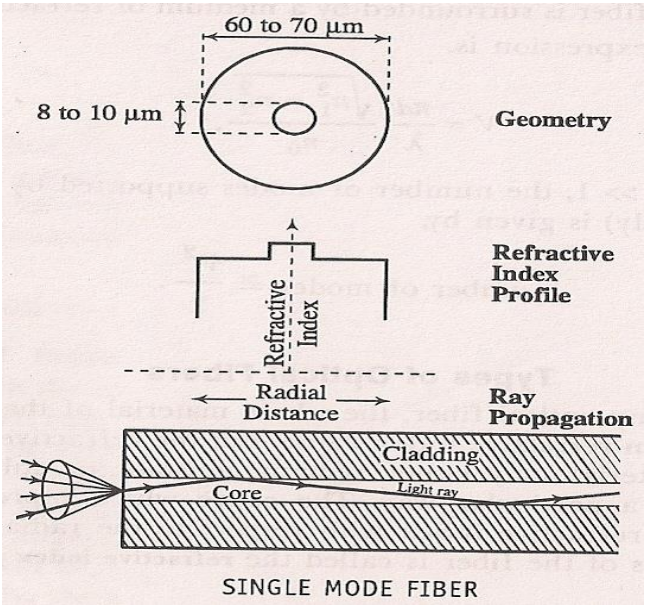
Refractive index profile :

The graph showing the variation of R.I with distance from axis to fibre is called R.I profile. There are 2 types of R.I profile, namely

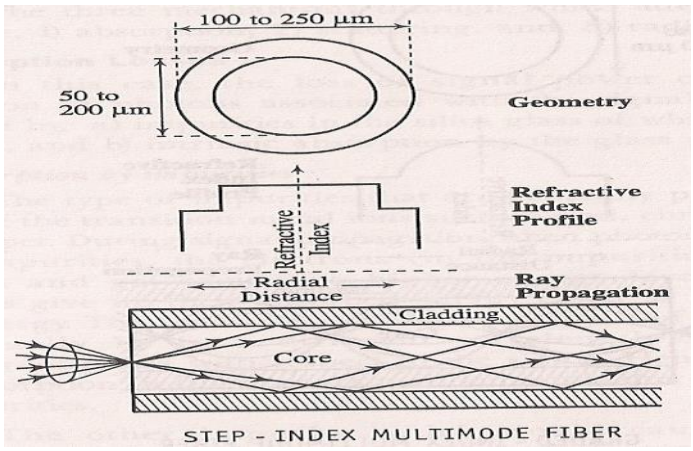
i) Step-index profile
ii) Graded-index profile.

In step index profile, R.I of core (n1) is uniform and constant throughout core of R.I. of cladding (n2) is uniform & constant throughout cladding [n1 > n2]. In graded index profile, R.I of core is maximum at the axis (n1), decreases uniformly on both sides reaches (n2) at the core cladding interface.

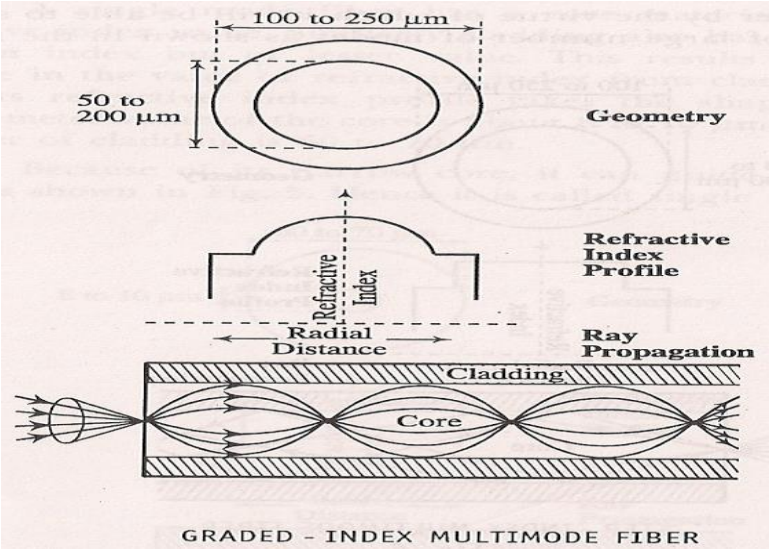
1) **Single mode fibre :** It has a core material of uniform refractive index. Similarly cladding also is made of a material whose refractive index is uniform
but less than that of core. So Refractive Index profile attains a stepwise shape. The core dia is 8 to 10 μm & cladding dia is 70 to 80 μm. Because of narrow
Core it can guide only a single mode. This is effectively used one and it needs lasers as source of light used in submarine cable system.



2) **Step-index multi mode fibre :** Similar to single mode fibre but core dia is 50 to 200 μm and cladding dia is 100 μm to 250 μm. It is capable of guiding a number of modes. In this also R.I profile has a stepwise shape. This accept LED or laser as source of light. Applied in data communication with low bandwidth requirement.



3) **Graded index multimode fibre (GRIN) :** The dia of core is 100 to 250 μm and that of cladding is 100 to 250 μm. The core material R.I decreases in radially outward direction from the axis and becomes equal to that of the cladding at the interface. It was either LED or laser as the source of light. Typical application in telephone trunk



Attenuation : It is the power loss suffered by optical signal, when it propagates through the fiber. It is also called fiber loss. It is represented by α.

$\alpha = -\frac{10}{L} log_{10}\left(\frac{p_o}{p_i}\right)$, where L → length of fiber in km

The unit of α is dB/km. Attenuation can be due to i) Absorption, ii) Scattering iii) Radiation.

Absorption : The power can be lost by intrinsic or extrinsic absorption. Each fiber material has its own nature of absorbing certain λ. This is called intrinsic absorption. The absorption by the impurities present in the fiber is called extrinsic absorption. Eg. : Hydroxy ions.

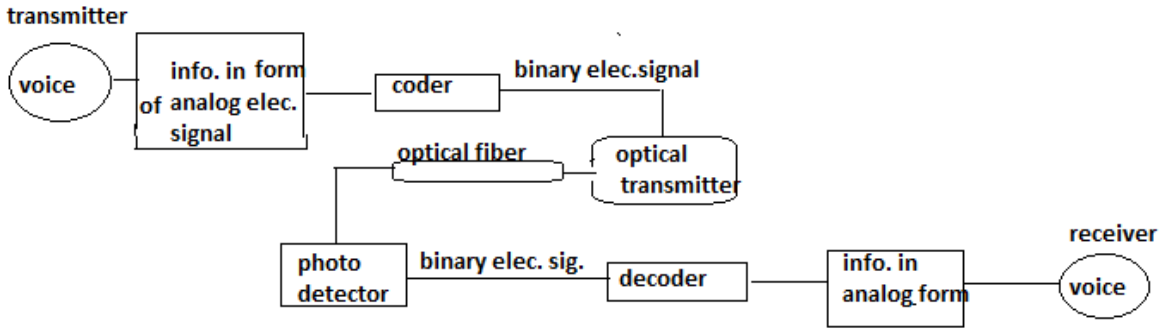
Scattering : The signal gets scattered by impurities of very small size. This scattering is more if λ of light is small, as scattering is proportional to $\frac{1}{\lambda^4}$. Due to very small variation in R.I. also, scattering occurs. The structural inhomogenities can also cause scattering.

Radiation: This occurs due to bending of fiber. It can be macroscopic and microscopic. The loss is negligible for macroscopic bends, microscopic bends are mainly manufacturing defects, due to which the I.R gets lost and the power loss occurs.

Application of Optical fibers

A. Point to point telecommunication system:

They are used in point to point telecommunication system. The block diagram is shown below.



The voice gets converted into analog electrical signal which gets coded in the binary form by coder. These are converted into optical pulses and get modulated by suitable light source. This is done by optical transmitter. From this the optical power is fed into optical fiber which gets transmitted through it by undergoing TIR. The output from optical fiber is fed into a photo detector which converts them into binary electrical signal. They are then decoded and then get converted into sound energy.

While getting transmitted through optical fiber the signal gets attenuated and the power drops. To regain the power a repeater unit, which has a receiver & transmitter kept side by side is used. A receiver receives the optical signal, converts into electrical signal, amplifies and modulates it and sends it to the transmitter. A transmitter amplifies the signal, converts it into optical signal and feed it into the fiber.